Examiners' Report
Summer 2026

Pearson Edexcel GCE

In Mathematics (9MA0)

Paper 02 Pure Mathematics

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General Comments

Overall, the 9MA0/02 paper provided plenty of challenges for the full cohort of learners.

It was often the case that learners were successful in the first 5 questions with 3(d) usually offering the first challenge and in some cases, learners did not gain marks in 4(b) where a calculator should not be used. The longer, later questions provided suitable challenge, whilst also providing restart opportunities for those who struggled with earlier parts of questions. There were also a range of independent marks that learners could access.

There were particular instances where learners did not gain marks:

- use of a calculator when the question included an instruction that a calculator should not be used e.g. 4(b)
- answers not given as requested e.g. the "equation of the model" in 5(a), 8(a) and 9(a)

These points aside, there were many succinct and elegant solutions to the more demanding problems towards the end of the paper.

Comments on individual questions

Question 1

This was answered well with the factor theorem correctly applied. The question specifically required the use of the factor theorem so any attempts to use long division or synthetic division were awarded no marks.

Most learners substituted $x = 2$ in order to set up a linear equation in k. A small number of learners used $x = -2$. The right-hand side of the equation was often omitted, and its presence could only be implied by subsequent working.

The most common error following substitution was the incorrect expansion of brackets leading to sign errors in the equation for k. There were also sign slips in the rearrangement of a simple linear equation of the form $ax + b = 0$ meaning the final accuracy mark was not gained following otherwise correct work.

Question 2

This question was generally well answered, with most learners making good progress through both parts of the question.

Part (a)

In part (a), the majority of learners were able to calculate h correctly and apply the trapezium rule to estimate the integral. The most common error seen in this part was incorrectly calculating the value of h by dividing the interval into 5 strips instead of 4, giving $h = 0.04$. This was most commonly seen when learners were using the formula $h = \frac{b-a}{n}$ as stated in the formula booklet, rather than working directly from the values in the table. The other incorrect value of h commonly seen was $h = 0.5$. Many learners were able to score the method mark for correctly applying the trapezium rule with their value of h, however a small number of learners made bracketing errors with their substitution, and these were rarely recovered to imply a correct method. Some learners mistakenly used the x values in their formula rather than the y values, and occasional arithmetic slips also prevented otherwise correct methods from receiving full credit.

Part (b)

In part (b), most learners integrated correctly and recognised the need to combine the integral with their value from part (a). However, a common error was to combine these values incorrectly, often integrating their answer to part (a) as part of their method rather than adding it on afterwards. A small number of learners omitted the addition of their answer from (a) altogether or substituted 0.8 and 0.6 directly into the given function. Learners who used prematurely rounded figures often reached an answer of 4.57, so the final accuracy mark was not gained. The follow through mark here meant many learners were still able to gain credit for their work despite errors in part (a).

Question 3

In this question on arithmetic series, most learners scored well in the first three parts. The fourth part proved more challenging, with many attempts seen that did not have a clear solution strategy.

Part (a)

In part (a), a correct summation formula in terms of a, d and n was used with the correct value of $n = 65$ in order to obtain the printed result in the majority of responses. Learners would be well advised to state any formula first before attempting to use it. Errors seen included missing brackets in the formula and a confusion between the summation formula and the n^{th} term formula.

Part (b)

In part (b), most learners achieved a relevant second equation in first term, *a*, and common difference, *d*, using the formula for the *n*th term of an arithmetic series. There were occasional uses of $a + nd$ rather than $a + (n - 1)d$ and, even rarer, a misread of the question leading to the use of the summation formula.

Part (c)

The solution of the pair of simultaneous equations in *a* and *d* required in part (c) was achieved by elimination, substitution or using a calculator. The use of a calculator was allowed in this part of the question as the solution did not rely entirely on calculator technology.

Part (d)

In part (d) it was essential to see working to demonstrate a convincing strategy as solutions relying entirely on calculator technology were not allowed. This warning was heeded as it was rare to see just an answer given.

The commonest method attempted for part (d) was to realise that the maximum sum would occur immediately before the terms of the series become negative. This led usually to the use of $a + (n - 1)d = 0$ with the found values of *a* and *d* to find a value for the number of terms *n*.

The solution of the equation was non-integer and so an integer value of *n* had to be found in order to proceed. There were instances of the non-integer value being used in the summation formula which received no credit.

Another method attempted was finding the maximum of the sum of *n* terms by using calculus or completing the square. Also seen was taking the sum as a quadratic in *n* and then finding the non-zero root. However, those choosing this route often just used the non-zero root as their value of *n* rather than realising that the maximum of the quadratic occurred halfway between the two roots. Rarely seen was the method of trial and improvement. This was acceptable provided that there was sufficient convincing working shown.

Question 4

This question was well attempted, with a large number of learners being awarded all of the marks in part (a) and at least some of the marks in part (b).

Part (a)

In part (a) most learners were very successful, expanding and simplifying the required expression and showing sufficient working to achieve the given result. Some learners approach the problem without writing the intermediate stage before proceeding to the final answer that is needed for a "show that" question and so were unable to gain the final accuracy mark. Algebraic slips were seen only a small number of responses; when seen, this was normally achieving a quadratic expression rather than a quartic from expanding their brackets.

Part (b)

Fewer learners scored full marks in part (b) than in part (a). Whilst most learners were able to substitute into the given expression correctly and form a quadratic in k^2, common errors included the incorrect expansion of brackets or failure to collect terms before their attempt at solving. Occasionally learners did not set their quadratic equal to zero before solving and simply factorised the expression on the left-hand side of their equation. There was evidence that many learners relied solely on calculator technology to solve their resulting quadratic equation, despite the requirement in the question to demonstrate their use of a non-calculator method, which meant that the final method and accuracy marks were not gained. Of those learners who attempted to solve the equation as it is or those who used a substitution, many just wrote down the solutions from their calculator, while others attempted to work backwards into an invalid factorisation for their expression. Some of those who did show their non-calculator method, did not gain marks for not finding both values of k or for not ruling out the negative value for k^2.

Question 5

Overall, this question was answered well by the majority of learners. Most learners demonstrated a secure understanding of the sinusoidal model and were able to use the given information effectively to determine the required parameters and interpret the model in context.

Part (a)

Most learners were able to score well on this part of the question. Most learners successfully used the given conditions to form simultaneous equations, solved them correctly to find the values of a and b, and then stated a complete equation for the model.

Learners who did not achieve full marks generally fell into one of a small number of categories. The most common issue was failing to write down the complete model after finding the correct values of a and b, despite this being explicitly requested. Such learners typically earned both method marks but did not gain the final accuracy mark. This omission occasionally caused difficulties later in the question, as learners proceeded directly to calculations in part (c) without ever having stated a complete model.

A further common error was the formation of correct simultaneous equations followed by errors in solving them, leading to incorrect values of a and/or b. Less frequently, learners substituted the given information incorrectly into the model, producing incorrect equations from the outset. A small number of learners worked in radians rather than degrees, which prevented them from obtaining the correct model despite otherwise sound methods.

Part (b)

This part was also answered successfully by most learners. The majority correctly interpreted *a* as the depth of water at midnight, the initial depth of water, or equivalently the depth of water when $t = 0$

A small proportion of responses identified *a* as the average or mean depth of water. Whilst less common, such responses were seen regularly. The most common incorrect response was to describe *a* as the minimum depth of water. Some learners gave contradictory statements, such as referring to the "initial minimum depth", whilst a small number provided answers with little or no reference to the context of the question or left the part unanswered.

Occasionally, learners incorrectly referred to the volume of water rather than its depth, indicating some confusion between the modelled quantity and the physical situation.

Part (c)

The method mark on this part was achieved by almost all learners, including many who had been unsuccessful in part (a). Most correctly attempted to use their model by substituting $t = 10.5$ to calculate a value for D. Even where the model itself was incorrect, this usually demonstrated sufficient understanding to gain the method mark.

A common error amongst those who did not gain the method mark was substituting $t = 10.3$, suggesting that time was incorrectly expressed in decimal hours. Other unsuccessful attempts either substituted correctly but did not evaluate the model or did not make any meaningful attempt at the question.

Very few learners chose alternative valid approaches such as substituting both $D = 4.9$ and $t = 10.5$, showing that the two sides were approximately equal. Very few learners worked backwards from $D = 4.9$ to find a corresponding value of t.

The accuracy mark was usually scored by those with a correct model. Most successfully demonstrated that their calculated value was sufficiently close to 4.9 and made an appropriate judgement regarding the suitability of the model. Some learners went beyond what was required by calculating percentage errors or using other numerical measures of comparison. Whilst unnecessary, such approaches were generally successful when applied correctly.

The most common reason for the final mark not to be awarded was having an incorrect model from part (a). Some learners with a correct value for *D* either made an incorrect conclusion about the validity of the model, failed to make any conclusion at all, or provided flawed numerical reasoning, which meant the final marks were not awarded. A notable misconception was the belief that the model must produce exactly the value given in the question in order to be valid. Stronger responses recognised that mathematical models are approximations and evaluated the model based on the closeness of the predicted value rather than exact agreement.

Overall, the question was accessible to the vast majority of learners. The key differentiators were the accurate completion of the model in part (a), correct interpretation of parameters in context, and an appreciation in part (c) that the validity of a model is judged by how closely its predictions match reality.

Question 6

This was a well-accessed question, with the vast majority of learners attempting it and scoring well. Most learners were familiar with the differentiation from first principles method and were able to set up the solution correctly, with very few blank responses.

The most successful responses followed the standard first principles method systematically, including an appropriate limiting statement and clear algebraic working. Learners who used the binomial expansion (or Pascal's Triangle) to expand $(x+h)^3$ generally made fewer errors than those who expanded using repeated multiplication with triple brackets. Showing each stage of the algebra clearly, rather than skipping intermediate steps, often helped learners avoid accuracy errors.

Common errors:

- errors when expanding the cubic, particularly when using triple brackets instead of binomial expansion
- arithmetic mistakes when multiplying $5(x^3 + 3x^2h + 3xh^2 + h^3)$ incorrectly with some terms not being multiplied by 5
- starting with $(5x+h)^3$ or expanding $(x+h)^3$ incorrectly
- cancelling the h terms by crossing them out without writing a subsequent line of working showing the simplified expression
- omitting derivative notation such as $f'(x)$ or $\frac{\mathrm{d}y}{\mathrm{d}x}$, despite otherwise obtaining the correct expression
- occasional errors with the coefficient 5 remaining incorrectly attached to x within the bracket

Overall, this was a well-attempted question, where marks were not awarded because of algebraic inaccuracies rather than a lack of understanding of the method.

Question 7

This question was generally answered well. Few learners left this question blank.

Most learners correctly found $\overrightarrow{BC}$ by subtraction although sign errors were fairly common. A frequent misconception was to add $\overrightarrow{AB}$ and $\overrightarrow{AC}$, giving $7\mathbf{i} + 2\mathbf{j} + \mathbf{k}$.

Almost all learners successfully used Pythagoras to find the length of at least one side of the triangle. Those using the cosine rule were sometimes prone to errors, including incorrect recall of the formula, incorrect substitution, or algebraic mistakes when rearranging. Some learners found the wrong angle, but most gave their final answer in degrees to one decimal place as instructed.

Many learners used the scalar product, which generally led to efficient and accurate solutions. Although the cross product was a valid alternative, it was not seen.

Question 8

Learners generally found parts (a) and (b) of this question to be accessible and full marks were quite common. Part (c) was more demanding, particularly in the requirement for providing a suitable reason for the answer.

Part (a)

In (a) most learners knew the correct approach and used $M = 25$ with $t = 0$ in the given equation. Many successfully reached the correct value of $k = 3$. Arithmetic slips when solving for k were rarely seen. A number of learners, however, did not write down the complete equation of the model, $M = \frac{200}{5+3e^{-0.15t}}$, as required by the question, so the final mark was not awarded.

Part (b)

In (b) most learners started correctly and used their model with $M = 35$ and attempted to find the value of t. Most learners reached an equation such as $105\mathrm{e}^{-0.15T} = 25$ or $\mathrm{e}^{-0.15T} = \frac{5}{21}$. Learners then generally knew the correct process to follow and used correct log work leading to the answer of $T = 9.57$ minutes. There were instances of incorrect log or exponent work, and a few learners showed insufficient working, probably have used the equation solver on their calculator, which was not permitted in this question. Credit was sometimes lost due to algebraic slips in rearranging, errors when taking the natural logarithms, for example mishandling coefficients or treating $\ln 3e^{-0.15t}$ incorrectly, or misreading values from part (a). Learners who made an error in part (a) were often still able to score the method marks here. Generally, this part was answered well.

Part (c)

In part (c) most learners correctly identified the upper limit $L = 40$ by considering the behaviour as t → ∞. The majority recognised that the exponential term tends to zero, but the quality of reasoning varied. Many provided a clear and rigorous limiting argument and scored well. Precise arguments with regard to $\mathrm{e}^{-0.15t}$ approaching zero as $t \longrightarrow \infty$ were rewarded. Learners who linked their reasoning clearly to the exponential term were often able to gain full marks. For the explanation, learners were required to have a numerical answer for the limit in the interval [39.9, 40] and then give a reason that gave the sense that as t became large, the exponential term was small. Often one of these parts was missing. A significant minority were not awarded the reasoning mark by attempting to substitute in large values of t, or by making incomplete references to asymptotes. A common error was to state e tends to zero. There were a few unsuccessful context-based explanations such as "the reaction will stop at some point", that did not address the mathematics of the model.

Question 9

Overall, this question was accessible for most learners, and the majority were able to gain marks, with many achieving high scores. Learners who laid out their work clearly and labelled each part of the question were more likely to access all available marks. Most errors arose not from a lack of understanding of the modelling process itself, but from issues of precision, communication, and attention to detail, particularly regarding units and contextual interpretation.

Part (a)

In part (a), most learners were able to use a correct strategy to obtain values or numerical expressions for *a* and *b*, with the most common approach being to take logarithms base 10 of both sides of the given equation and to compare this with a linear form. Most learners recognised that $\log a = 0.304$ and $\log b = 0.0108$ and in the main this then led to correct values. A common issue was that some learners did not state the complete equation for L after finding these values and therefore did not have access to the final accuracy mark. There were occasional errors with logs such as using base e instead of base 10. Misreads of both 0.0108 and 0.304 were quite common. Not reading the 3 decimal place requirement and instead giving 3 significant figures was quite often seen. Other methods, such as finding a value for *a* and then using a coordinate on the line to find a value for *b*, tended to be less successful.

Part (b)

In part (b), many learners correctly interpreted *a* as the initial amount of coal mined, often referencing the year 1700. However, a significant number of responses did not gain the mark due to a lack of precision with units. The omission of "millions" or "tonnes", or both, was very common. A few learners gave incorrect interpretations, such as describing *a* as a rate of change, or the minimum amount of coal.

Part (c)

Part (c) was generally well answered with most learners recognising the need to substitute $t = 300$ into their model. However, errors from part (a) resulted in incorrect final answers. As in part (b), omission of units prevented some learners from gaining the accuracy mark. A small number of learners misunderstood the question and attempted to find the difference in the amount of coal mined between 2000 and 1999. Although not required, some learners attempted to convert their answer into millions by multiplying by 10^6 or into billions, which in general was done correctly.

Part (d)

Responses to part (d) were more varied. Many learners correctly identified that the estimate was unreliable due to extrapolation beyond the range of the data, which was the most common valid justification. Some also referred to contextual factors, such as changes in energy use or the development of alternative energy sources. However, a number of responses were too vague to be credited, for example simply stating that the answer was “unreliable” without justification, or giving a justification that was not relevant, such as stating that coal would run out, or that the value obtained was too large, without linking this to the validity of the model.

Key points

- Write down the complete equation for the model.
- Note accuracy requirements carefully, particularly the distinction between decimal places and significant figures.
- Give answers with the correct units.
- Explicitly state whether the model is reliable and provide a clear reason.
- Predictions involve extrapolation significantly beyond the data range.

Question 10

Part (a)

Almost all learners attempted part (a) of the question, and many gained full marks. The most popular method was to multiply through by $(1-x)(1-2x)$ and then use $x=1, 0.5$ and 0 to achieve values for *A*, *B* and *C*. A few made arithmetical errors when calculating either *B* or *C* and then also had an incorrect value for *A*. Some learners compared coefficients with mixed success. Those that used long division to find *A* were split between those using the remainder, $5x-1$ to find *B* and *C* against those mistakenly still using $4x^2-x+1$. A small number of learners attempted to divide $4x^2 - x + 1$ by either $x - 1$ or $1 - 2x$ but in the majority of cases, did not give a correct denominator of their remainder was and so did not have a correct method.

Part (b)

In part (b) most learners realised that they had to write the denominator of the fractions as a power of -1 and then went on to find the binomial expansions. The majority had 3 correct terms, but less successful responses showed errors, particularly sign slips, when simplifying their expansions. Almost all learners that had tried the expansions were able to gain the mark for combining like terms. A notable number of learners resorted to the original expression and proceeded with expanding $(4\mathrm{x}^2 - x + 1)(1 - x)^{-1}(1 - 2x)^{-1}$. This approach was condoned and was more difficult as it required better organisation of the multiplication to include all the relevant terms. These responses had mixed success with some terms missing. A small number of learners used a method not on the specification, namely Maclaurin series. Those who opted for this method were usually successful.

Part (c)

Part (c) was more challenging, but a good proportion of learners gave either $|x| < \frac{1}{2}$ or $-\frac{1}{2} < x < \frac{1}{2}$ as the final answer. Several only stated $x < 0.5$. Some learners who did use the modulus sign couldn't apply it correctly, leading to incorrect answers such as $|x| < -\frac{1}{2}$. Others gave a domain for both expansions and failed to select the most restrictive case.

Question 11

Performance on Question 11 showed a clear divide between part (a), which most learners handled well, and part (b), which proved highly discriminating.

Part (a)

Responses to this part were generally strong, with many demonstrating a secure understanding of integration by substitution. Most responses correctly applied the chain rule to find $\frac{\mathrm{d}u}{\mathrm{d}x}$, although a common error involved omitting the factor of 5, resulting in $\frac{du}{dx} = \frac{1}{2}(5x-2)^{-\frac{1}{2}}$ instead of the correct $\frac{5}{2}(5x-2)^{-\frac{1}{2}}$. Transforming this derivative to express d*x* in terms of d*u* was widely achieved.

Rearranging the initial substitution to make *x* the subject followed by finding $\frac{\mathrm{d}x}{\mathrm{d}u}$ was the most popular and effective approach, which frequently resulted in fewer algebraic errors than methods beginning with $\frac{\mathrm{d}u}{\mathrm{d}x}$. Where errors did occur, they often involved a coefficient of $\frac{5}{2}$ instead of $\frac{2}{5}$ or mistakenly rewriting the $(5x-2)^{\frac{1}{2}}$ as $u^{\frac{1}{2}}$ rather than *u*.

Some responses show some difficulties with algebraic rearrangement during the substitution stage, occasionally treating the distinct expressions $\frac{2}{5u}$ and $\frac{2}{5}u$ as equivalent.

The process of changing the substitution limits from *x* values to *u* values was completed accurately by most learners, independent of their success in establishing the algebraic substitution. Stronger responses maintained excellent clarity throughout these steps.

Most common errors were:

- wrong coefficient, frequently $k = \frac{5}{2}$ from erroneous rearranging
- errors in dealing with the negative power in the rearranging, resulting in *u* being in the denominator, leading to (*u* + 3) in their final integral which was then moved to the numerator with no explanation

- forgetting to include the d*u* in their final answer

Clear step by step working was often lacking, causing errors when dealing with reciprocals.

Part (b)

This part proved to be more challenging, but responses that made an appropriate initial step typically progressed well. A prevalent misconception was to assume that $\frac{u}{u+3}$ could be integrated directly via a reverse chain rule to an expression such as $u\,ln(u+3)$. Some learners benefitted from employing a secondary substitution such as $z = u + 3$.

Those using the algebraic division method generally integrated correctly to obtain $u - 3\ln(u+3)$. Those who chose the secondary substitution often successfully achieved $z - 3\ln z$, though some responses did not adjust the integration limits to 4 and 8.

Processing of the logarithms was generally sound but minor slips occasionally occurred when converting ln 8 and ln 4 to terms of ln 2. This sometimes resulted in an incorrect final sign in the answer e.g. $\frac{8}{5}+\frac{6}{5}\ln 2$. Also a notable number of responses omitted the constant factor of $\frac{2}{5}$ from part (a).

Alternative strategies, such as integration by parts were significantly less successful. This approach required a demanding double application of the technique, which was rarely completed successfully.

Question 12

Part (a)

Part (a) was accessible to most learners with nearly all obtaining two correct coordinates for substituting in the value of *t* to find *x* and *y* coordinates.

Part (b)

In part (b), learners could often score all four method marks even if they made accuracy errors. Learners were generally confident with knowing the general method for parametric differentiation and attempted to find derivatives of both *y* and *x* with respect to *t*. As a result, virtually all learners used the method they were directed to in the question, rather than attempting a Cartesian method.

Finding $\frac{\mathrm{d}y}{\mathrm{d}t}$ was the most challenging aspect for the first method mark due to the double angle and product rule. Some responses did not recognise the need to apply these rules and where this was the case we generally saw them simply differentiating each of the trig functions giving $\frac{dy}{dt} = 2\cos t \cos 2t$. It was reasonably common for learners to make sign or coefficient slips here. A number of learners also replaced the double angle with single angles, and this led to a couple of variations in the form for $\frac{\mathrm{d}y}{\mathrm{d}t}$ and with more manipulation came more potential for slips.

Finding $\frac{\mathrm{d}x}{\mathrm{d}t}$ was generally done more successfully as it was a simpler function to deal with for the second method mark. If an error was made here, it was either a sign error or to keep the 1 +… not realising the first part differentiated to zero.

The third method mark was often scored for using $t = \frac{\pi}{4}$ with $\pm\frac{\mathrm{d}y}{\mathrm{d}x}$ or $\pm\frac{\mathrm{d}x}{\mathrm{d}y}$. A noticeable minority of learners unnecessarily tried to simplify their fraction before substituting $t = \frac{\pi}{4}$, which often led to an incorrect result. The fourth method mark was a dependent method mark, but it was only dependent on the previous method mark, which meant that incorrect differentiation did not prevent learners from obtaining this mark. If mistakes were made it was for using the negative gradient rather than the negative reciprocal. Some did not attempt the normal gradient, simply using the tangent gradient. Occasionally learners lost the final accuracy mark for not giving their answer in the required form, with *x* and *y* terms on the same side, meaning that even where the accuracy marks were not awarded, the method marks were available.

Part (c)

Part (c) was noticeably more challenging. The first method mark was obtained by most learners usually by eliminating *y*, rather than *x*. However very few realised that $\left(\sqrt{2}y\right)^2$ would give them $2y^2$ straight away and became involved in extended expansion of brackets with roots. Manipulating squared brackets involving surds did mean that the first accuracy mark was not awarded to some learners. It was quite common for learners to score only the first or first two marks on this question for a few reasons. If they didn't obtain the final accuracy mark in (b) then generally the accuracy marks were not available in (c) (unless it was lost due to just having the wrong form). Some learners just reached a cubic equation but did not continue to solve it, or they used a calculator to solve their cubic. The demand for the second method mark was that they used the fact that $(x-1)$ was a factor, so we needed to see them use the factor theorem or find the remaining quadratic factor by some other method such as inspection or polynomial long division. It wasn't common to see learners equate coefficients here. This method mark was accessible even if learners had the wrong cubic equation, as long as they fulfilled the basic criteria for finding the remaining quadratic factor. For learners who obtained a correct cubic and showed a valid method to find *x*, most went on to successfully obtain the correct exact value.

Question 13

This question assessed learners' ability to carry out implicit differentiation, manipulate algebraic expressions involving $\frac{dy}{dx}$, determine the gradient of a normal, and complete an exact surd calculation without the use of calculator technology.

While part (a) was generally well attempted with many learners demonstrating confidence in routine implicit differentiation, responses to part (b) were noticeably weaker. A significant proportion of responses did not gain full marks in the second part due to a combination of misreading the question (finding the tangent instead of the normal), incorrect algebraic handling of reciprocals, and an over-reliance on calculator technology despite explicit instructions to the contrary.

Part (a)

This part was generally completed well, with many learners scoring well. Most made a successful start by differentiating the y^2 term correctly, demonstrating a sound understanding of the core concept of implicit differentiation.

Common Errors:

- The most frequent error occurred when differentiating the $-xy$ term. Some learners did not recognise the need for the product rule entirely, differentiating only the *x* or the *y*. Others recognised the product rule but struggled with the negative sign, leading to sign errors. Learners who formally labelled their terms $u, v, \frac{du}{dx}, \frac{dv}{dx}$ were generally much more successful in avoiding these mistakes.
- A notable number of learners managed the complex *y* terms but did not differentiate the simpler elements of the equation. Common slips included failing to differentiate the $-x$ term correctly or leaving the constant 79 in the

differentiated equation instead of reducing it to zero.

- Learners who differentiated correctly generally had little difficulty collecting the $\frac{\mathrm{d}y}{\mathrm{d}x}$ terms and factorising to make it the subject. However, less successful responses occasionally featured sign errors when moving terms across the equals sign. A few learners included a $\frac{\mathrm{d}y}{\mathrm{d}x} =$ at the very start of their working, though this rarely interfered with their subsequent rearrangement.

Part (b)

Performance on part (b) was very varied. This part required learners to substitute given surd coordinates into their derivative, find the negative reciprocal, and rationalise the denominator algebraically. Very few learners achieved full marks here and those who did often found the negative reciprocal of their general expression first before substituting the complex surd coordinates. This approach minimized arithmetic errors and provided a clearer path to the final rationalisation.

Common Errors:

- Many learners simply substituted the coordinates to evaluate the gradient of the tangent and then stopped, making no attempt to find the gradient of the normal. Such responses were not awarded marks for this section.
- Among those who did attempt to find the normal gradient, algebraic errors were prevalent. Many simply took the reciprocal without making it negative or incorrectly multiplied both the numerator and denominator by -1. A particular algebraic misconception saw learners split their $\frac{\mathrm{d}y}{\mathrm{d}x}$ fraction (which had a single denominator) into two separate fractions and then take the negative reciprocal of each fraction individually.
- The question contained an explicit instruction that all stages of working should be shown and that solutions relying on calculator technology were not acceptable. Despite this, a significant number of learners who reached

the correct unsimplified normal gradient simply used their calculators to jump straight to the final answer in the required $a + b\sqrt{2}$ form. Because they failed to show the intermediate step of multiplying the numerator and denominator by the conjugate surd, they were not awarded the final two marks.

Question 14

Part (a)

The first part of the question was generally well attempted, with many learners successfully picking up the method mark for differentiating an exponential function of the form a^{kx}. Most learners correctly identified that the derivative involved both the original exponential term and a natural logarithm factor, arriving at $6^{-3x} \times (-3 \ln 6)$ or an equivalent form. Where marks were not awarded in this part, it was commonly due to structural errors in applying the chain rule. A notable number of learners attempted to apply a polynomial power rule to the exponential function, reducing the power by one to give derivatives involving terms like 6^{-3x-1}. Another common error was the total omission of the scaling factor $\ln 6$, with learners writing $\frac{dy}{dx} = -3 \times 6^{-3x}$. There was also an algebraic misconception where learners attempted to multiply the scalar result of the chain rule directly into the base of the exponential term, writing $\frac{\mathrm{d}y}{\mathrm{d}x} = 18^{-3x} \times \ln 6$. Where learners showed a correct unsimplified expression first, full marks were awarded in part (a) under the principle of ignoring subsequent working. However, learners who wrote down $18^{-3x} \times \ln 6$ directly with no intermediate steps were not awarded marks. A small number of learners chose to change the base of the function to base e or used implicit differentiation before differentiating. While these alternative approaches are fully acceptable and were credited when completed successfully, they were often lengthier with minor processing errors.

Part (b)

Learners found part (b) more challenging, and it was not uncommon to see this part left blank if an incorrect or incomplete derivative had been found in part (a). To access the method mark here, learners needed to demonstrate a clear strategy to isolate x from the exponent by correctly introducing logarithms to both sides of their equation. Minor rearrangement slips were condoned at this stage, provided the logarithmic step was structurally sound. A primary reason for the final accuracy mark being withheld was not meeting the demand for an exact value. Many learners utilised their calculators to evaluate decimal approximations for intermediate terms, such as writing $\ln 6 \approx 1.791$ which meant the final mark was not available. Weaknesses in negative sign tracking was also a common issue. Learners frequently made processing slips when separating their variables, often leading to invalid mathematical operations such as attempting to take the logarithm of a negative value. A minority of learners did not recognise the relationship between the algebraic derivative found in part (a) and the equation to be solved in part (b). Instead of setting their algebraic expression for $\frac{\mathrm{d}y}{\mathrm{d}x}$ equal to -1 and solving for x, these learners chose $x = -1$ and substituted it directly into their derivative expression from part (a). Because learners applied index and logarithmic laws in a variety of different sequences, an array of fully correct equivalent exact forms for *x* were seen, for example:

$$\frac{1}{3\ln 6}\ln(3\ln 6),\ \frac{1}{3}\log_6(\ln 216),\ \log_{216}(\ln 216),\quad -\frac{1}{3}\log_6\left(\frac{1}{\ln 216}\right),\ \frac{\ln(3\ln 6)}{\ln 216},\quad \frac{\ln 3+\ln(\ln 6)}{3\ln 6}$$

Question 15

Question 15 comprised of a circle equation where learners needed to find the centre and radius and for part (b), they need to find where a line with a known gradient intersected the circle at two distinct points.

Part (a)

Part (a) was answered well, with most completing the square successfully and obtaining the centre and radius of the circle. In less successful responses, the coefficients of x or y were not halved when completing the square. Other errors included reversing the coordinates of the centre, giving $(3, -2)$ rather than $(-3, 2)$ and not accounting correctly for the constant terms introduced when completing the square, sometimes leading to a radius of 7.

Part (b)

In part (b), most made progress by substituting $y = 2x + k$ into either the original equation of the circle or the completed-square form found in part (a). Some responses using the completed-square form encountered errors when expanding the brackets or collecting like terms and arithmetic errors during the rearrangement often resulted in an incorrect three-term quadratic. In particular, some learners lost the "= 75" in the course of their expanding and simplifying and ultimately replaced it with an "= 0". Most who obtained a quadratic equation recognised that the discriminant was needed to determine when the line and circle intersected at two distinct points.

Many responses obtained two critical values for k. The final presentation of the range was less consistent. Common issues included selecting the regions outside the two critical values, using a union symbol rather than the interval between the values, or not giving the answer in set notation as requested. Stronger responses used the condition $b^2 - 4ac > 0$ and gave the final result as $\{k; 8 - 5\sqrt{15} < k < 8 + 5\sqrt{15}\}$. The equivalent interval notation, although acceptable, was rarely seen.

A small number of learners attempted alternative geometrical methods, including implicit differentiation and consideration of the radial line, although these approaches were generally less successful than direct substitution and use of the discriminant.